

Theorem S4 Suppose that Assumptions 1-2 and S3-S4 hold. Then,

$$\beta_{fe}^X = E \left[\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t}^X \Delta_{g,t} \right].$$

Theorem S4 shows that under a modified version of the common trends assumption accounting for the covariates, β_{fe}^X identifies a weighted sum of the $\Delta_{g,t}^{TR}$ s, as β_{fe} in Theorem 1, with different but still potentially negative weights.⁷ Assumption S4 may be more plausible than Assumption 5, but adding covariates may increase the prevalence of negative weights, or the correlation between the weights and the $\Delta_{g,t}$ s, thus making β_{fe}^X less robust to heterogeneous effects than β_{fe} .

3.4 2SLS regressions

Researchers have sometimes estimated 2SLS versions of Regressions 1 and 2. Our main conclusions also apply to those regressions. Let $\hat{\beta}_{fe}^{2SLS}$ denote the coefficient of $D_{i,g,t}$ in a 2SLS regression of $Y_{g,t}$ on group and period fixed effects and $D_{i,g,t}$, using a variable $Z_{g,t}$ constant within each group $\times$ period as the instrument for $D_{i,g,t}$. $Z_{g,t}$ typically represents an incentive for treatment allocated at the group $\times$ period level. For instance, Duflo (2001) studies the effect of years of schooling on wages in Indonesia, using a primary school construction program as an instrument. Specifically, she estimates a 2SLS regression of wages on cohort and district of birth fixed effects and years of schooling, using the interaction of belonging to a cohort entering primary school after the program was completed and the number of schools constructed in one's district of birth as the instrument for years of schooling.

Remark that $\hat{\beta}_{fe}^{2SLS} = \hat{\beta}_{fe}^Y / \hat{\beta}_{fe}^D$, where $\hat{\beta}_{fe}^Y$ (resp. $\hat{\beta}_{fe}^D$) is the coefficient of $Z_{g,t}$ in the reduced-form regression of $Y_{g,t}$ (resp. the first-stage regression of $D_{g,t}$) on group and period fixed effects and $Z_{g,t}$. Then let $\beta_{fe}^{2SLS} = E[\hat{\beta}_{fe}^Y] / E[\hat{\beta}_{fe}^D]$.⁸ Following Imbens and Angrist (1994), for any $z \in \text{Supp}(Z)$ let $D_{i,g,t}(z)$ denote the potential treatment of unit i in (g, t) if $Z_{i,g,t} = z$. It follows from Theorem 1 that under a common trends assumption on $D_{i,g,t}(0)$, $E[\hat{\beta}_{fe}^D]$ is equal to a weighted sum of the average effects of the instrument on the treatment in each group and time period, with potentially many negative weights. Similarly, under a common trends assumption

⁷In a previous version of this paper, we had shown that under a different, and arguably less natural, common trends assumption, β_{fe}^X identifies a weighted sum of the $\Delta_{g,t}^{TR}$, with the same weights as in Theorem 1. We thank an anonymous referee for pointing out issues with the common trends assumption we had previously proposed.

⁸We do not consider here $E[\hat{\beta}_{fe}^{2SLS}]$, as the 2SLS estimator may not have an expectation. Moreover, under conditions similar to those imposed in Section 5 of the paper, β_{fe}^{2SLS} is the probability limit of $\hat{\beta}_{fe}^{2SLS}$, which makes β_{fe}^{2SLS} the proper estimand here.

on $Y_{i,g,t}(D_{i,g,t}(0))$ instead of $Y_{i,g,t}(0)$, $E[\widehat{\beta}_{fe}^Y]$ is equal to a weighted sum of the average effects of the instrument on the outcome, again with potentially many negative weights. For instance, in Duflo (2001), under a common trends assumption on $D_{i,g,t}(0)$, the number of years of schooling individuals would complete if zero new schools were constructed in their district, the first stage coefficient identifies a weighted sum of the effect of one new school on years of schooling in every district, with many negative weights.⁹

Hence, it is only if the average effects of $Z_{g,t}$ on $Y_{i,g,t}$ and $D_{i,g,t}$ are constant across groups and periods, or if the weights are uncorrelated to treatment effects as in Assumption 7, that the reduced-form and first-stage coefficients respectively identify the average effect of $Z_{i,g,t}$ on $Y_{i,g,t}$ and $D_{i,g,t}$. Then, this implies that β_{fe}^{2SLS} identifies, under the conditions in Imbens and Angrist (1994), the LATE of $D_{i,g,t}$ on $Y_{i,g,t}$ among units that comply with the instrument.¹⁰

4 Extending the DID_M estimator

Theorem 3 can be extended to the case where $D_{i,g,t}$ is not binary but takes values in $\mathcal{D} = \{0, \dots, \bar{d}\}$. The causal effect we consider is the switchers' causal response

$$\delta^{SCR} = E \left(\frac{1}{N_{D,S}} \sum_{(i,g,t): t \geq 2, D_{g,t} \neq D_{g,t-1}} [Y_{i,g,t}(\max(D_{g,t}, D_{g,t-1})) - Y_{i,g,t}(\min(D_{g,t}, D_{g,t-1}))] \right),$$

where $N_{D,S} = \sum_{(g,t): t \geq 2} N_{g,t} |D_{g,t} - D_{g,t-1}|$. Note that $\delta^{SCR} = \delta^S$ when $D_{i,g,t}$ is binary.

We identify δ^{SCR} under the following two conditions, which generalize Assumptions 4-5 and 9-12 to non-binary treatments.

Assumption S5 (*Mean independence between a group's outcome and other groups treatments*)
For all $(d, g, t) \in \mathcal{D} \times \{1, \dots, G\} \times \{1, \dots, T\}$, $E(Y_{g,t}(d) | \mathbf{D}) = E(Y_{g,t}(d) | \mathbf{D}_g)$.

Assumption S6 (*Strong exogeneity*) For all $(d, g, t) \in \mathcal{D} \times \{1, \dots, G\} \times \{2, \dots, T\}$, $E(Y_{g,t}(d) - Y_{g,t-1}(d) | \mathbf{D}_g) = E(Y_{g,t}(d) - Y_{g,t-1}(d))$.

Assumption S7 (*Common trends*) For every d , for all $t \geq 2$ and g , $E(Y_{g,t}(d) - Y_{g,t-1}(d))$ does not vary across g .

⁹New schools were constructed in every district, so this application falls into the heterogeneous adoption case.

¹⁰In the special case with two groups and two periods, a binary incentive for treatment, and where only group 1 in period 1 receives the incentive, de Chaisemartin (2010) and Hudson et al. (2015) show that in a 2SLS regression of $Y_{i,g,t}$ on $1\{g = 2\}$, $1\{t = 2\}$ and $D_{i,g,t}$, using $Z_{g,t} = 1\{g = 2\}1\{t = 2\}$ as the instrument, the coefficient of $D_{i,g,t}$ identifies a LATE under common trends assumptions on $Y_{i,g,t}(D_{i,g,t}(0))$ and $D_{i,g,t}(0)$. However, the discussion above shows that this result does not generalize to applications with multiple groups and periods or a non-binary instrument, as in Duflo (2001) where the number of new schools constructed varies across districts.

Assumption S8 (*Existence of “stable” groups*) For all $t \in \{2, \dots, T\}$, for all $(d, d') \in \mathcal{D}^2$, $d \neq d'$, if there is at least one $g \in \{1, \dots, G\}$ such that $D_{g,t-1} = d$ and $D_{g,t} = d'$, then there exists at least one $g' \neq g$, $g' \in \{1, \dots, G\}$ such that $D_{g',t-1} = D_{g',t} = d$.

When the treatment takes a large number of values, Assumption S8 may be violated. A solution, then, is to consider a modified treatment variable $\tilde{D}_{g,t} = h(D_{g,t})$ that groups together several values of $D_{g,t}$, to ensure that Assumption S8 holds for $\tilde{D}_{g,t}$. For instance, if the treatment can be equal to 0, 1, 2, or 3, and there is a group whose treatment switches from 2 to 3 between periods 1 and 2, but no group whose treatment remains equal to 2 between those two dates, one may define $\tilde{D}_{g,t} = \min(D_{g,t}, 2)$ if there is a group whose treatment is equal to 3 at periods 1 and 2. Then, Theorem S5 below still holds, after replacing $D_{g,t}$ by $\tilde{D}_{g,t}$ in the DID $_{d,d',t}$ estimators defined below, and if Assumption S7 is replaced by the requirement that $E(Y_{g,t}(d) - Y_{g,t-1}(d))$ only depends on t and $h(d)$.

In order to define DID $_M$ in this context, let us introduce, for all $(d, d', t) \in \mathcal{D}^2 \times \{2, \dots, T\}$,

$$\begin{aligned} \text{DID}_{d,d',t} = [1\{d < d'\} - 1\{d' < d\}] & \left[\sum_{(g,t): D_{g,t}=d', D_{g,t-1}=d, t \geq 2} \frac{N_{g,t}}{N_{d,d',t}} [Y_{g,t} - Y_{g,t-1}] \right. \\ & \left. - \sum_{(g,t): D_{g,t}=D_{g,t-1}=d, t \geq 2} \frac{N_{g,t}}{N_{d,d,t}} [Y_{g,t} - Y_{g,t-1}] \right], \end{aligned}$$

where $N_{d,d',t}$ is defined as in (3) for any $(d, d') \in \mathcal{D}^2$. Then

$$\text{DID}_M = \sum_{t=2}^T \sum_{(d,d') \in \mathcal{D}^2, d \neq d'} \frac{N_{d,d',t}}{N_{D,S}} \text{DID}_{d,d',t}.$$

If the treatment is binary, the DID $_M$ estimator defined above is equal to that defined in Section 4 of the paper.

Theorem S5 Suppose that $D_{i,g,t} \in \mathcal{D}$ and Assumptions 1-2 and S5-S8 hold. Then $E[\text{DID}_M] = \delta^{SCR}$.

Theorem S5 generalizes Theorem 3 to non-binary treatments. We can also extend Theorem 4 in the same way to construct placebo tests of Assumption S7.

Finally, Theorem 3 can also be extended to the case with covariates. Under versions of Assumptions 10 and 11 written conditional on X , a conditional version of the DID $_M$ estimator is consistent for δ^S under the common support condition $\text{Supp}(X_{d,g,t}) = \text{Supp}(X)$. We refer to de Chaisemartin and D'Haultfœuille (2018) for further details.

5 Statistical properties of DID_M and inference on δ^S

In this section, we establish the asymptotic properties of DID_M and construct confidence intervals on δ^S based on DID_M. We consider an asymptotic framework where the number of groups G tends to infinity. To define the confidence intervals, let $P_{d,d',t} = N_{d,d',t}/G$ and

$$Q_{d,d',t} = \frac{1}{G} \sum_g N_{g,t} \mathbb{1}\{D_{g,t} = d, D_{g,t-1} = d'\} (Y_{g,t} - Y_{g,t-1}).$$

Then, let $\hat{\sigma}^2 = \sum_g \hat{\psi}_g^2 / G$, with

$$\begin{aligned} \hat{\psi}_g &= \frac{G}{N_S} \sum_{t>1} N_{g,t} \left[\mathbb{1}\{D_{g,t} \neq D_{g,t-1}\} (Y_{g,t} - Y_{g,t-1} - \text{DID}_M) - \hat{\psi}_{g,t}^B \right], \\ \hat{\psi}_{g,t}^B &= \frac{1}{P_{0,0,t}} \left[\mathbb{1}\{D_{g,t} > D_{g,t-1}\} Q_{0,0,t} + P_{1,0,t} \mathbb{1}\{D_{g,t} = D_{g,t-1} = 0\} \left(Y_{g,t} - Y_{g,t-1} - \frac{Q_{0,0,t}}{P_{0,0,t}} \right) \right] \\ &\quad + \frac{1}{P_{1,1,t}} \left[\mathbb{1}\{D_{g,t} < D_{g,t-1}\} Q_{1,1,t} + P_{0,1,t} \mathbb{1}\{D_{g,t} = D_{g,t-1} = 1\} \left(Y_{g,t} - Y_{g,t-1} - \frac{Q_{1,1,t}}{P_{1,1,t}} \right) \right]. \end{aligned}$$

We consider confidence intervals of the form

$$\text{CI}_{1-\alpha}(\delta^S) = \left[\text{DID}_M - z_{1-\alpha/2} \frac{\hat{\sigma}}{\sqrt{G}}, \text{DID}_M + z_{1-\alpha/2} \frac{\hat{\sigma}}{\sqrt{G}} \right],$$

where $z_{1-\alpha/2}$ denotes the quantile of order $1 - \alpha/2$ of a standard normal variable.

We now establish the asymptotic properties of DID_M and $\text{CI}_{1-\alpha}(\delta^S)$ under the following assumptions. Hereafter, we denote $\mathbf{U} = (P_{0,0,1}, Q_{0,0,1}, \dots, P_{1,1,T}, Q_{1,1,T})$.

Assumption S9 (*Existence of moments and limits*) $\sup_{g,t} N_{g,t} < +\infty$ and $\sup_{(d,g,t)} E(Y_{g,t}^4(d)) < +\infty$. $\lim_G E[\mathbf{U}]$ and $\lim_G G \times V(\mathbf{U})$ exist.

Assumption S10 (*Positive probability of “stable” groups and existence of switchers*) For all $(d, g, t) \in \{0, 1\} \times (\mathbb{N} \setminus \{0\}) \times \{2, \dots, T\}$, $\Pr(D_{g,t} = 1-d, D_{g,t-1} = d) > 0$ implies $\lim_G E[P_{1-d,d,t}] > 0$ and $\lim_G E[P_{d,d,t}] > 0$. Moreover, $\lim_G E[P_{0,1,t} + P_{1,0,t}] > 0$ for at least one t .

Assumption S9 imposes the (uniform) existence of moments of order 4 of $Y_{g,t}(d)$, and that some non-random averages converge as G tends to infinity. These assumptions ensure that we can apply law of large numbers and central limit theorems in our set-up where groups are independent but not necessarily identically distributed. Assumption S10 imposes that when at least one group switches from d to $1-d$ with a positive probability on a given period, then on average over all groups, the limit probabilities of switching from d to $1-d$ will be positive as $G \rightarrow \infty$. The limit probability of remaining at d will also be positive. This latter condition may be seen as a weaker version of Assumption 11, as it imposes the existence of “stable” groups only